In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
#Load dataset
df = pd.read_csv(r"C:/Users/salun/OneDrive/Desktop/service_requests.csv",low_mem
```

In [3]:
```
df.head()
```

Out[3]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Loc
0	32310363	12/31/2015 11:59:45 PM	01-01-2016 00:55	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Stre
1	32309934	12/31/2015 11:59:44 PM	01-01-2016 01:26	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
2	32309159	12/31/2015 11:59:29 PM	01-01-2016 04:51	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
3	32305098	12/31/2015 11:57:46 PM	01-01-2016 07:43	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Stre
4	32306529	12/31/2015 11:56:58 PM	01-01-2016 03:24	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Stre

5 rows × 53 columns

In [4]:
```
# visualize Dataset
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  int64
 1   Created Date                    364558 non-null  str
 2   Closed Date                     362177 non-null  str
 3   Agency                          364558 non-null  str
 4   Agency Name                     364558 non-null  str
 5   Complaint Type                  364558 non-null  str
 6   Descriptor                      358057 non-null  str
 7   Location Type                   364425 non-null  str
 8   Incident Zip                    361560 non-null  float64
 9   Incident Address                312859 non-null  str
 10  Street Name                     312859 non-null  str
 11  Cross Street 1                  307370 non-null  str
 12  Cross Street 2                  306753 non-null  str
 13  Intersection Street 1           51120 non-null   str
 14  Intersection Street 2           50512 non-null   str
 15  Address Type                    361306 non-null  str
 16  City                            361561 non-null  str
 17  Landmark                        375 non-null     str
 18  Facility Type                   362169 non-null  str
 19  Status                          364558 non-null  str
 20  Due Date                        364555 non-null  str
 21  Resolution Description          364558 non-null  str
 22  Resolution Action Updated Date  362156 non-null  str
 23  Community Board                 364558 non-null  str
 24  Borough                         364558 non-null  str
 25  X Coordinate (State Plane)      360528 non-null  float64
 26  Y Coordinate (State Plane)      360528 non-null  float64
 27  Park Facility Name              364558 non-null  str
 28  Park Borough                    364558 non-null  str
 29  School Name                     364558 non-null  str
 30  School Number                   364558 non-null  str
 31  School Region                   364557 non-null  str
 32  School Code                     364557 non-null  str
 33  School Phone Number             364558 non-null  str
 34  School Address                  364558 non-null  str
 35  School City                     364558 non-null  str
 36  School State                    364558 non-null  str
 37  School Zip                      364557 non-null  str
 38  School Not Found                364558 non-null  str
 39  School or Citywide Complaint    0 non-null       float64
 40  Vehicle Type                    0 non-null       float64
 41  Taxi Company Borough            0 non-null       float64
 42  Taxi Pick Up Location           0 non-null       float64
 43  Bridge Highway Name             297 non-null     str
 44  Bridge Highway Direction        297 non-null     str
 45  Road Ramp                       262 non-null     str
 46  Bridge Highway Segment          262 non-null     str
 47  Garage Lot Name                 0 non-null       float64
 48  Ferry Direction                 1 non-null       str
 49  Ferry Terminal Name             2 non-null       str
 50  Latitude                        360528 non-null  float64
 51  Longitude                       360528 non-null  float64
 52  Location                        360528 non-null  str
dtypes: float64(10), int64(1), str(42)
memory usage: 147.4 MB
```

In [5]:
```
# Columns of Dataframe
print(df.columns)
```

```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='str')
```

In [6]:
```
#Identify the shape of the dataset
df.shape
```

Out[6]:
```
(364558, 53)
```

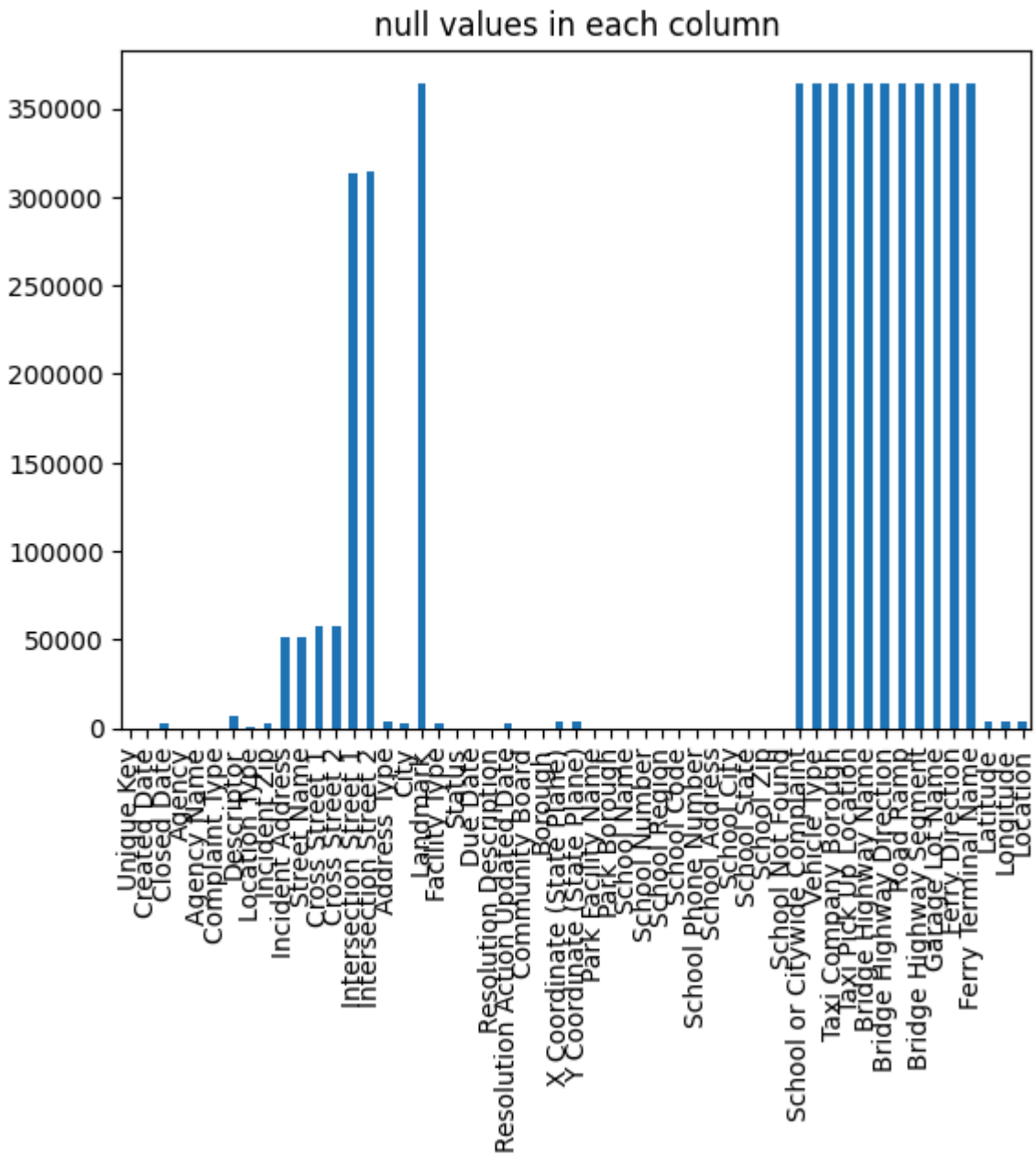
In [7]:
```
# Null values
df.isnull().sum()
```

Out[7]:
```
Unique Key                          0
Created Date                        0
Closed Date                      2381
Agency                              0
Agency Name                         0
Complaint Type                      0
Descriptor                       6501
Location Type                     133
Incident Zip                     2998
Incident Address                51699
Street Name                     51699
Cross Street 1                  57188
Cross Street 2                  57805
Intersection Street 1          313438
Intersection Street 2          314046
Address Type                     3252
City                             2997
Landmark                       364183
Facility Type                    2389
Status                              0
Due Date                            3
Resolution Description              0
Resolution Action Updated Date   2402
Community Board                     0
Borough                             0
X Coordinate (State Plane)       4030
Y Coordinate (State Plane)       4030
Park Facility Name                  0
Park Borough                        0
School Name                         0
School Number                       0
School Region                       1
School Code                         1
School Phone Number                 0
School Address                      0
School City                         0
School State                        0
School Zip                          1
School Not Found                    0
School or Citywide Complaint   364558
Vehicle Type                   364558
Taxi Company Borough           364558
Taxi Pick Up Location          364558
Bridge Highway Name            364261
Bridge Highway Direction       364261
Road Ramp                      364296
Bridge Highway Segment         364296
Garage Lot Name                364558
Ferry Direction                364557
Ferry Terminal Name            364556
Latitude                         4030
Longitude                        4030
Location                         4030
dtype: int64
```

Perform basic data exploratory analysis:

In [8]:
```
# Frequency plot of null values
df.isnull().sum().plot(kind ='bar')
```

```
Figesize=(25,10)
plt.title("null values in each column")
plt.show()
```



In [9]:
```
# Remove records where closed data is null
df = df[df['Closed Date'].notnull()]
print(df)
```

```
        Unique Key            Created Date       Closed Date Agency  \
0         32310363  12/31/2015 11:59:45 PM  01-01-2016 00:55   NYPD
1         32309934  12/31/2015 11:59:44 PM  01-01-2016 01:26   NYPD
2         32309159  12/31/2015 11:59:29 PM  01-01-2016 04:51   NYPD
3         32305098  12/31/2015 11:57:46 PM  01-01-2016 07:43   NYPD
4         32306529  12/31/2015 11:56:58 PM  01-01-2016 03:24   NYPD
...            ...                     ...                ...    ...
364553    29609918        01-01-2015 00:04  01-01-2015 10:22   NYPD
364554    29608392        01-01-2015 00:04  01-01-2015 02:25   NYPD
364555    29607589        01-01-2015 00:01  01-01-2015 00:20   NYPD
364556    29610889        01-01-2015 00:01  01-01-2015 02:42   NYPD
364557    29611816        01-01-2015 00:00  01-01-2015 02:47   NYPD

                            Agency Name           Complaint Type  \
0       New York City Police Department  Noise - Street/Sidewalk
1       New York City Police Department         Blocked Driveway
2       New York City Police Department         Blocked Driveway
3       New York City Police Department          Illegal Parking
4       New York City Police Department          Illegal Parking
...                                 ...                      ...
364553  New York City Police Department          Illegal Parking
364554  New York City Police Department          Noise - Vehicle
364555  New York City Police Department  Noise - Street/Sidewalk
364556  New York City Police Department         Blocked Driveway
364557  New York City Police Department         Blocked Driveway

                         Descriptor    Location Type  Incident Zip  \
0                  Loud Music/Party  Street/Sidewalk       10034.0
1                         No Access  Street/Sidewalk       11105.0
2                         No Access  Street/Sidewalk       10458.0
3       Commercial Overnight Parking  Street/Sidewalk       10461.0
4                  Blocked Sidewalk  Street/Sidewalk       11373.0
...                             ...              ...           ...
364553               Blocked Hydrant  Street/Sidewalk       11421.0
364554                Car/Truck Horn  Street/Sidewalk       10468.0
364555             Loud Music/Party  Street/Sidewalk       10031.0
364556                    No Access  Street/Sidewalk       10466.0
364557                    No Access  Street/Sidewalk       11420.0

             Incident Address  ... Bridge Highway Name  \
0         71 VERMILYEA AVENUE  ...                 NaN
1             27-07 23 AVENUE  ...                 NaN
2       2897 VALENTINE AVENUE  ...                 NaN
3         2940 BAISLEY AVENUE  ...                 NaN
4               87-14 57 ROAD  ...                 NaN
...                       ...  ...                 ...
364553          84-25 85 ROAD  ...                 NaN
364554   2555 SEDGWICK AVENUE  ...                 NaN
364555    508 WEST 139 STREET  ...                 NaN
364556    931 EAST 226 STREET  ...                 NaN
364557      123-19 135 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
0                           NaN       NaN                    NaN
1                           NaN       NaN                    NaN
2                           NaN       NaN                    NaN
3                           NaN       NaN                    NaN
4                           NaN       NaN                    NaN
...                         ...       ...                    ...
364553                      NaN       NaN                    NaN
```

```
364554                      NaN       NaN                     NaN
364555                      NaN       NaN                     NaN
364556                      NaN       NaN                     NaN
364557                      NaN       NaN                     NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
0                  NaN             NaN                 NaN  40.865682
1                  NaN             NaN                 NaN  40.775945
2                  NaN             NaN                 NaN  40.870325
3                  NaN             NaN                 NaN  40.835994
4                  NaN             NaN                 NaN  40.733060
...                ...             ...                 ...        ...
364553             NaN             NaN                 NaN  40.695145
364554             NaN             NaN                 NaN  40.867830
364555             NaN             NaN                 NaN  40.821647
364556             NaN             NaN                 NaN  40.886361
364557             NaN             NaN                 NaN  40.674212

        Longitude                                 Location
0      -73.923501   (40.86568153633767, -73.92350095571744)
1      -73.915094  (40.775945312321085, -73.91509393898605)
2      -73.888525  (40.870324522111424, -73.88852464418646)
3      -73.828379   (40.83599404683083, -73.82837939584206)
4      -73.874170  (40.733059618956815, -73.87416975810375)
...           ...                                       ...
364553 -73.860949   (40.69514470265117, -73.86094888534394)
364554 -73.907178   (40.86782963689454, -73.90717786644662)
364555 -73.950873  (40.821646626438095, -73.95087342885292)
364556 -73.853290   (40.88636077906953, -73.85329048666742)
364557 -73.803585  (40.674211762243935, -73.80358548685278)

[362177 rows x 53 columns]
```

In [10]:
```
df['Created Date']
```

Out[10]:
```
0         12/31/2015 11:59:45 PM
1         12/31/2015 11:59:44 PM
2         12/31/2015 11:59:29 PM
3         12/31/2015 11:57:46 PM
4         12/31/2015 11:56:58 PM
                    ...
364553          01-01-2015 00:04
364554          01-01-2015 00:04
364555          01-01-2015 00:01
364556          01-01-2015 00:01
364557          01-01-2015 00:00
Name: Created Date, Length: 362177, dtype: str
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
Unique Key                          0
Created Date                        0
Closed Date                         0
Agency                              0
Agency Name                         0
Complaint Type                      0
Descriptor                       6496
Location Type                     130
Incident Zip                      675
Incident Address                51686
Street Name                     51686
Cross Street 1                  55331
Cross Street 2                  55464
Intersection Street 1          311549
Intersection Street 2          311673
Address Type                      929
City                              674
Landmark                       361802
Facility Type                      18
Status                              0
Due Date                            1
Resolution Description              0
Resolution Action Updated Date     39
Community Board                     0
Borough                             0
X Coordinate (State Plane)       1707
Y Coordinate (State Plane)       1707
Park Facility Name                  0
Park Borough                        0
School Name                         0
School Number                       0
School Region                       1
School Code                         1
School Phone Number                 0
School Address                      0
School City                         0
School State                        0
School Zip                          1
School Not Found                    0
School or Citywide Complaint   362177
Vehicle Type                   362177
Taxi Company Borough           362177
Taxi Pick Up Location          362177
Bridge Highway Name            361880
Bridge Highway Direction       361880
Road Ramp                      361915
Bridge Highway Segment         361915
Garage Lot Name                362177
Ferry Direction                362177
Ferry Terminal Name            362177
Latitude                         1707
Longitude                        1707
Location                         1707
dtype: int64
```

In [14]:
```python
# Analyze the date column, and remove entries that have an incorrect timeline
df['Created Date']
pd.to_datetime(df['Created Date'], errors='coerce')
```

```
df['Closed Date']
pd.to_datetime(df['Closed Date'], errors='coerce')
```

```
C:\Users\salun\AppData\Local\Temp\ipykernel_16344\1753231604.py:3: UserWarning: C
ould not infer format, so each element will be parsed individually, falling back
to `dateutil`. To ensure parsing is consistent and as-expected, please specify a
format.
  pd.to_datetime(df['Created Date'], errors='coerce')
```

Out[14]:

```
0        2016-01-01 00:55:00
1        2016-01-01 01:26:00
2        2016-01-01 04:51:00
3        2016-01-01 07:43:00
4        2016-01-01 03:24:00
                 ...
364553   2015-01-01 10:22:00
364554   2015-01-01 02:25:00
364555   2015-01-01 00:20:00
364556   2015-01-01 02:42:00
364557   2015-01-01 02:47:00
Name: Closed Date, Length: 362177, dtype: datetime64[us]
```

In [15]:

```
#time elapsed
df = df [df['Closed Date']>=
df['Created Date']]
print(df)
```

```
        Unique Key            Created Date             Closed Date Agency  \
26        32305916  12/31/2015 11:26:41 PM  12/31/2015 11:53:31 PM   NYPD
36        32306281  12/31/2015 11:13:47 PM  12/31/2015 11:32:32 PM   NYPD
38        32308014  12/31/2015 11:11:33 PM  12/31/2015 11:41:01 PM   NYPD
40        32307354  12/31/2015 11:11:18 PM  12/31/2015 11:16:13 PM   NYPD
41        32308746  12/31/2015 11:10:03 PM  12/31/2015 11:14:04 PM   NYPD
...            ...                     ...                     ...    ...
364553    29609918        01-01-2015 00:04        01-01-2015 10:22   NYPD
364554    29608392        01-01-2015 00:04        01-01-2015 02:25   NYPD
364555    29607589        01-01-2015 00:01        01-01-2015 00:20   NYPD
364556    29610889        01-01-2015 00:01        01-01-2015 02:42   NYPD
364557    29611816        01-01-2015 00:00        01-01-2015 02:47   NYPD

                            Agency Name            Complaint Type  \
26      New York City Police Department  Noise - House of Worship
36      New York City Police Department          Blocked Driveway
38      New York City Police Department   Noise - Street/Sidewalk
40      New York City Police Department        Noise - Commercial
41      New York City Police Department        Noise - Commercial
...                                 ...                       ...
364553  New York City Police Department           Illegal Parking
364554  New York City Police Department           Noise - Vehicle
364555  New York City Police Department   Noise - Street/Sidewalk
364556  New York City Police Department          Blocked Driveway
364557  New York City Police Department          Blocked Driveway

              Descriptor        Location Type  Incident Zip  \
26      Loud Music/Party     House of Worship       10031.0
36             No Access      Street/Sidewalk       11213.0
38      Loud Music/Party      Street/Sidewalk       11211.0
40      Loud Music/Party  Club/Bar/Restaurant       10017.0
41          Loud Talking     Store/Commercial       11216.0
...                  ...                  ...           ...
364553   Blocked Hydrant      Street/Sidewalk       11421.0
364554    Car/Truck Horn      Street/Sidewalk       10468.0
364555  Loud Music/Party      Street/Sidewalk       10031.0
364556         No Access      Street/Sidewalk       10466.0
364557         No Access      Street/Sidewalk       11420.0

            Incident Address  ... Bridge Highway Name  \
26       452 WEST 147 STREET  ...                 NaN
36       1285 ST JOHNS PLACE  ...                 NaN
38                       NaN  ...                 NaN
40              862 2 AVENUE  ...                 NaN
41          95 HALSEY STREET  ...                 NaN
...                      ...  ...                 ...
364553         84-25 85 ROAD  ...                 NaN
364554  2555 SEDGWICK AVENUE  ...                 NaN
364555   508 WEST 139 STREET  ...                 NaN
364556   931 EAST 226 STREET  ...                 NaN
364557     123-19 135 STREET  ...                 NaN

       Bridge Highway Direction Road Ramp Bridge Highway Segment  \
26                          NaN       NaN                    NaN
36                          NaN       NaN                    NaN
38                          NaN       NaN                    NaN
40                          NaN       NaN                    NaN
41                          NaN       NaN                    NaN
...                         ...       ...                    ...
364553                      NaN       NaN                    NaN
```

```
364554                      NaN       NaN                     NaN
364555                      NaN       NaN                     NaN
364556                      NaN       NaN                     NaN
364557                      NaN       NaN                     NaN

       Garage Lot Name Ferry Direction Ferry Terminal Name   Latitude  \
26                 NaN             NaN                 NaN  40.826102
36                 NaN             NaN                 NaN  40.670843
38                 NaN             NaN                 NaN  40.714007
40                 NaN             NaN                 NaN  40.752467
41                 NaN             NaN                 NaN  40.681870
...                ...             ...                 ...        ...
364553             NaN             NaN                 NaN  40.695145
364554             NaN             NaN                 NaN  40.867830
364555             NaN             NaN                 NaN  40.821647
364556             NaN             NaN                 NaN  40.886361
364557             NaN             NaN                 NaN  40.674212

        Longitude                                 Location
26     -73.945663     (40.826101709196, -73.94566339279595)
36     -73.935556  (40.670842567107634, -73.93555563656233)
38     -73.941715  (40.714007352909874, -73.94171469793936)
40     -73.970263  (40.752466784765815, -73.97026303855291)
41     -73.949372   (40.68186974794423, -73.94937196628798)
...           ...                                       ...
364553 -73.860949   (40.69514470265117, -73.86094888534394)
364554 -73.907178   (40.86782963689454, -73.90717786644662)
364555 -73.950873  (40.821646626438095, -73.95087342885292)
364556 -73.853290   (40.88636077906953, -73.85329048666742)
364557 -73.803585  (40.674211762243935, -73.80358548685278)

[321134 rows x 53 columns]
```

In [16]:

```python
# View the descriptive statistics for the newly created
df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
df
```

```
C:\Users\salun\AppData\Local\Temp\ipykernel_16344\746447771.py:2: UserWarning: Co
uld not infer format, so each element will be parsed individually, falling back t
o `dateutil`. To ensure parsing is consistent and as-expected, please specify a f
ormat.
  df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
C:\Users\salun\AppData\Local\Temp\ipykernel_16344\746447771.py:3: UserWarning: Co
uld not infer format, so each element will be parsed individually, falling back t
o `dateutil`. To ensure parsing is consistent and as-expected, please specify a f
ormat.
  df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

Out[16]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
26	32305916	2015-12-31 23:26:41	2015-12-31 23:53:31	NYPD	New York City Police Department	Noise - House of Worship	Loud Music/Party
36	32306281	2015-12-31 23:13:47	2015-12-31 23:32:32	NYPD	New York City Police Department	Blocked Driveway	No Access
38	32308014	2015-12-31 23:11:33	2015-12-31 23:41:01	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
40	32307354	2015-12-31 23:11:18	2015-12-31 23:16:13	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party
41	32308746	2015-12-31 23:10:03	2015-12-31 23:14:04	NYPD	New York City Police Department	Noise - Commercial	Loud Talking
...	...	...	...	...	...	...	...
364553	29609918	2015-01-01 00:04:00	2015-01-01 10:22:00	NYPD	New York City Police Department	Illegal Parking	Blocked Hydrant
364554	29608392	2015-01-01 00:04:00	2015-01-01 02:25:00	NYPD	New York City Police Department	Noise - Vehicle	Car/Truck Horn
364555	29607589	2015-01-01 00:01:00	2015-01-01 00:20:00	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
364556	29610889	2015-01-01 00:01:00	2015-01-01 02:42:00	NYPD	New York City Police Department	Blocked Driveway	No Access
364557	29611816	2015-01-01 00:00:00	2015-01-01 02:47:00	NYPD	New York City Police Department	Blocked Driveway	No Access

321134 rows × 53 columns

In [18]:

```
# convert into seconds
df['Request_Closing_Time'] = (df['Closed Date'] - df['Created Date']).dt.total_s
print(df)
```

```
        Unique Key        Created Date         Closed Date Agency  \
26        32305916 2015-12-31 23:26:41 2015-12-31 23:53:31   NYPD
36        32306281 2015-12-31 23:13:47 2015-12-31 23:32:32   NYPD
38        32308014 2015-12-31 23:11:33 2015-12-31 23:41:01   NYPD
40        32307354 2015-12-31 23:11:18 2015-12-31 23:16:13   NYPD
41        32308746 2015-12-31 23:10:03 2015-12-31 23:14:04   NYPD
...            ...                 ...                 ...    ...
364553    29609918 2015-01-01 00:04:00 2015-01-01 10:22:00   NYPD
364554    29608392 2015-01-01 00:04:00 2015-01-01 02:25:00   NYPD
364555    29607589 2015-01-01 00:01:00 2015-01-01 00:20:00   NYPD
364556    29610889 2015-01-01 00:01:00 2015-01-01 02:42:00   NYPD
364557    29611816 2015-01-01 00:00:00 2015-01-01 02:47:00   NYPD

                            Agency Name            Complaint Type  \
26      New York City Police Department  Noise - House of Worship
36      New York City Police Department          Blocked Driveway
38      New York City Police Department   Noise - Street/Sidewalk
40      New York City Police Department        Noise - Commercial
41      New York City Police Department        Noise - Commercial
...                                 ...                       ...
364553  New York City Police Department           Illegal Parking
364554  New York City Police Department           Noise - Vehicle
364555  New York City Police Department   Noise - Street/Sidewalk
364556  New York City Police Department          Blocked Driveway
364557  New York City Police Department          Blocked Driveway

              Descriptor        Location Type  Incident Zip  \
26      Loud Music/Party     House of Worship       10031.0
36             No Access      Street/Sidewalk       11213.0
38      Loud Music/Party      Street/Sidewalk       11211.0
40      Loud Music/Party  Club/Bar/Restaurant       10017.0
41          Loud Talking     Store/Commercial       11216.0
...                  ...                  ...           ...
364553   Blocked Hydrant      Street/Sidewalk       11421.0
364554    Car/Truck Horn      Street/Sidewalk       10468.0
364555  Loud Music/Party      Street/Sidewalk       10031.0
364556         No Access      Street/Sidewalk       10466.0
364557         No Access      Street/Sidewalk       11420.0

            Incident Address  ... Bridge Highway Direction Road Ramp  \
26       452 WEST 147 STREET  ...                      NaN       NaN
36       1285 ST JOHNS PLACE  ...                      NaN       NaN
38                       NaN  ...                      NaN       NaN
40              862 2 AVENUE  ...                      NaN       NaN
41          95 HALSEY STREET  ...                      NaN       NaN
...                      ...  ...                      ...       ...
364553         84-25 85 ROAD  ...                      NaN       NaN
364554  2555 SEDGWICK AVENUE  ...                      NaN       NaN
364555   508 WEST 139 STREET  ...                      NaN       NaN
364556   931 EAST 226 STREET  ...                      NaN       NaN
364557     123-19 135 STREET  ...                      NaN       NaN

       Bridge Highway Segment Garage Lot Name Ferry Direction  \
26                        NaN             NaN             NaN
36                        NaN             NaN             NaN
38                        NaN             NaN             NaN
40                        NaN             NaN             NaN
41                        NaN             NaN             NaN
...                       ...             ...             ...
364553                    NaN             NaN             NaN
```

```
364554                    NaN             NaN             NaN
364555                    NaN             NaN             NaN
364556                    NaN             NaN             NaN
364557                    NaN             NaN             NaN

       Ferry Terminal Name   Latitude  Longitude  \
26                     NaN  40.826102 -73.945663
36                     NaN  40.670843 -73.935556
38                     NaN  40.714007 -73.941715
40                     NaN  40.752467 -73.970263
41                     NaN  40.681870 -73.949372
...                    ...        ...        ...
364553                 NaN  40.695145 -73.860949
364554                 NaN  40.867830 -73.907178
364555                 NaN  40.821647 -73.950873
364556                 NaN  40.886361 -73.853290
364557                 NaN  40.674212 -73.803585

                                       Location Request_Closing_Time
26         (40.826101709196, -73.94566339279595)               1610.0
36      (40.670842567107634, -73.93555563656233)               1125.0
38      (40.714007352909874, -73.94171469793936)               1768.0
40      (40.752466784765815, -73.97026303855291)                295.0
41       (40.68186974794423, -73.94937196628798)                241.0
...                                           ...                  ...
364553   (40.69514470265117, -73.86094888534394)              37080.0
364554   (40.86782963689454, -73.90717786644662)               8460.0
364555  (40.821646626438095, -73.95087342885292)               1140.0
364556   (40.88636077906953, -73.85329048666742)               9660.0
364557  (40.674211762243935, -73.80358548685278)              10020.0

[321134 rows x 54 columns]
```

In [19]:
```
# View the descriptive statistics for the newly created column
df["Request_Closing_Time"].describe()
```

Out[19]:
```
count    3.211340e+05
mean     1.486686e+04
std      2.207162e+04
min      6.000000e+01
25%      4.200000e+03
50%      8.940000e+03
75%      1.794000e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

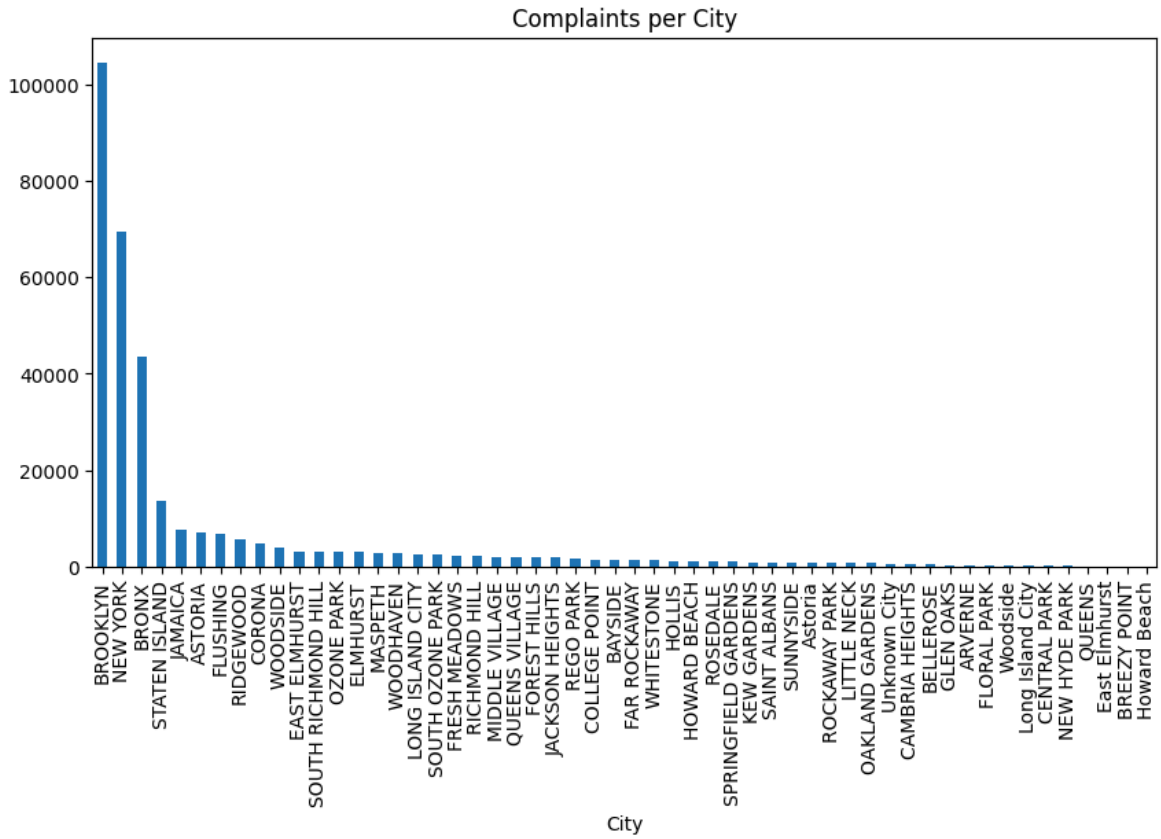
In [20]:
```
# Null valuses in complaint type & city
df[['Complaint Type', 'City']].isnull().sum()
```

Out[20]:
```
Complaint Type      0
City              559
dtype: int64
```

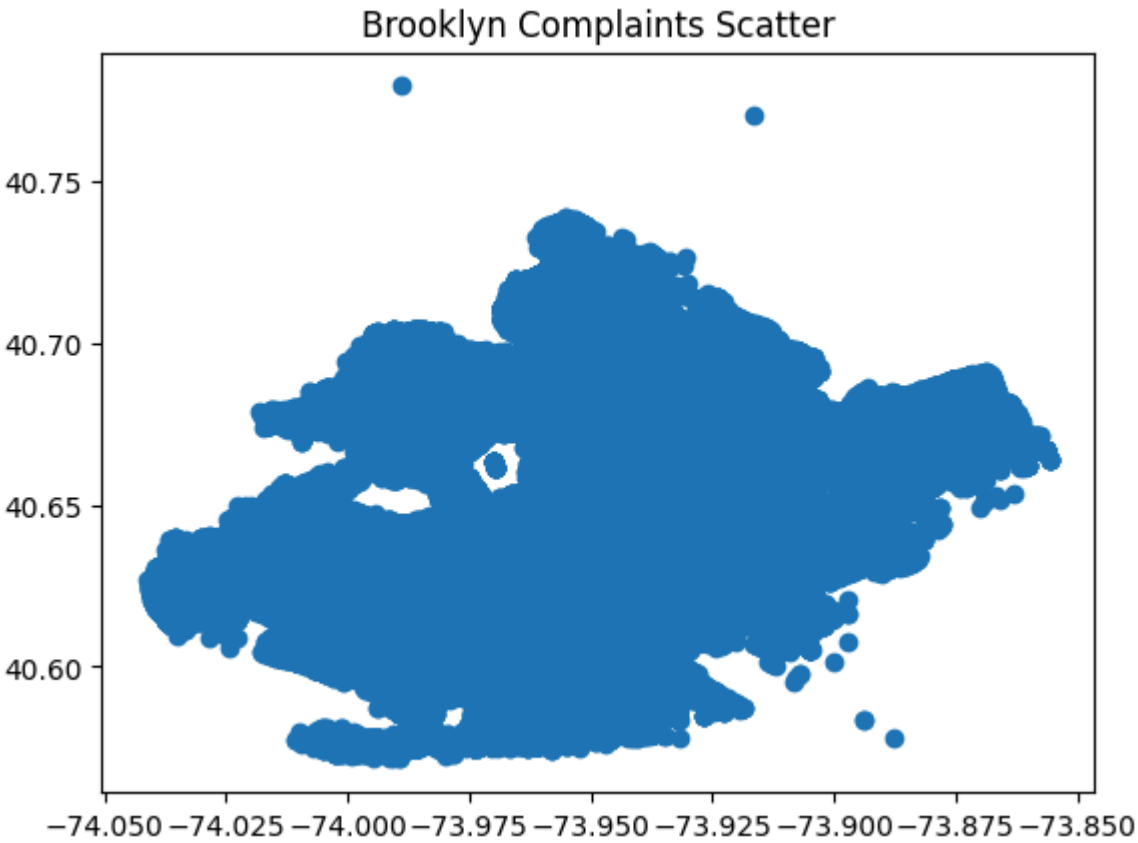
In [21]:
```
# Impute the NA value with Unknown City
print(df[['Complaint Type','City']].isnull().sum())

df['City'] = df['City'].fillna("Unknown City")

print(df['City'].isnull().sum())
```

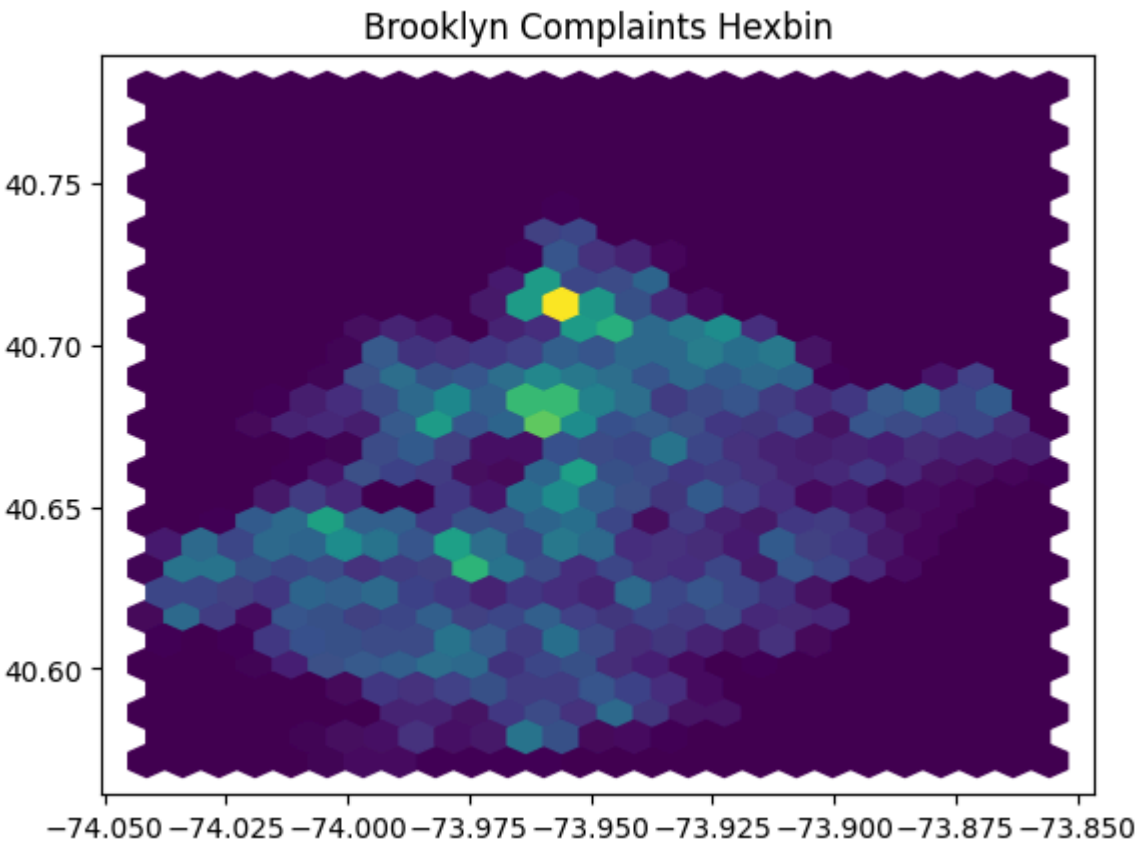
```
Complaint Type      0
City              559
dtype: int64
0
```

In [22]:
```
# Draw a frequency plot for the complaints in each city
df['City'].value_counts().plot(kind='bar', figsize=(10,5))
plt.title("Complaints per City")
plt.show()
```



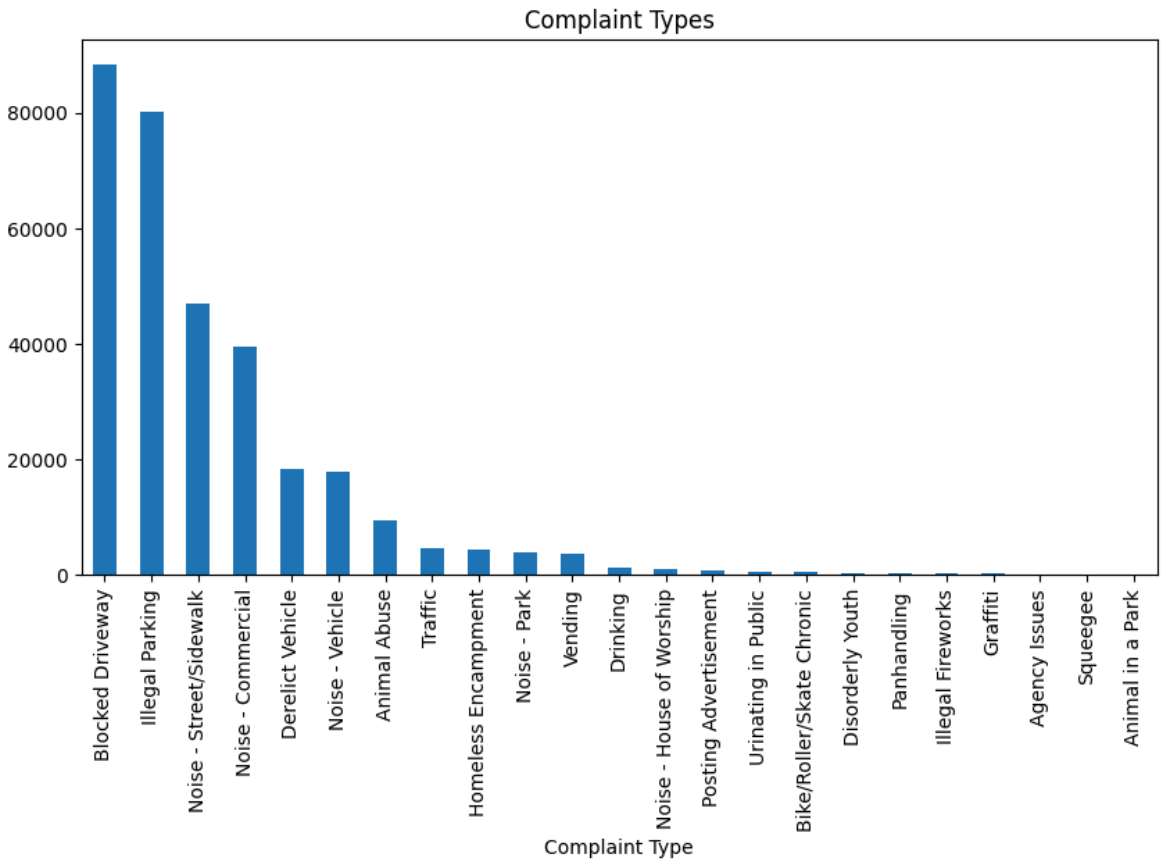
In [23]:
```
# Create a scatter and hexbin plot of the concentration of complaints across Bro
brooklyn = df[df['City'] == 'BROOKLYN']
plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'])
plt.title("Brooklyn Complaints Scatter")
plt.show()
plt.hexbin(brooklyn['Longitude'], brooklyn['Latitude'], gridsize=25)
plt.title("Brooklyn Complaints Hexbin")
plt.show()
```





Major Types of complaints

In [24]:
```
# Plot a bar graph to show the types of complaints
df['Complaint Type'].value_counts().plot(kind='bar', figsize=(10,5))
plt.title("Complaint Types")
plt.show()
```



In [25]:
```
#Check the frequency of various types of complaints for New York City
nyc = df[df['City'] == 'NEW YORK']
nyc['Complaint Type'].value_counts()
```

Out[25]:
```
Complaint Type
Noise - Street/Sidewalk      20386
Noise - Commercial           17028
Illegal Parking              12618
Noise - Vehicle               5780
Homeless Encampment           2672
Blocked Driveway              2374
Vending                       2224
Animal Abuse                  1723
Traffic                       1617
Noise - Park                  1171
Derelict Vehicle               577
Drinking                       291
Urinating in Public            232
Bike/Roller/Skate Chronic      228
Noise - House of Worship       203
Panhandling                    178
Disorderly Youth                78
Posting Advertisement           43
Illegal Fireworks               37
Graffiti                        20
Squeegee                         4
Name: count, dtype: int64
```

In [26]:
```
#Find the top 10 complaint types
Top10 = df['Complaint Type'].value_counts().head(10)
print(Top10)
```

```
Complaint Type
Blocked Driveway           88239
Illegal Parking            80040
Noise - Street/Sidewalk    46966
Noise - Commercial         39452
Derelict Vehicle           18395
Noise - Vehicle            17730
Animal Abuse                9305
Traffic                     4654
Homeless Encampment         4218
Noise - Park                3807
Name: count, dtype: int64
```

In [27]:

```
#Display the various types of complaints in each city
pd.crosstab(df['City'], df['Complaint Type'])
```

Out[27]:

Complaint Type City	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorder You
ARVERNE	0	43	0	0	43	29	
ASTORIA	0	146	0	16	3001	374	
Astoria	0	0	0	0	141	12	
BAYSIDE	0	50	0	0	461	194	
BELLEROSE	0	15	0	1	126	109	
BREEZY POINT	0	2	0	0	3	1	
BRONX	0	1715	0	21	14935	2066	
BROOKLYN	0	2833	0	110	31784	5276	
CAMBRIA HEIGHTS	0	14	0	0	156	128	
CENTRAL PARK	0	0	0	0	0	0	
COLLEGE POINT	0	31	0	0	534	198	
CORONA	0	93	0	0	3230	62	
EAST ELMHURST	0	78	0	1	1688	120	
ELMHURST	0	51	0	2	1761	82	
East Elmhurst	0	0	0	0	0	1	
FAR ROCKAWAY	0	96	0	0	347	187	
FLORAL PARK	0	7	0	0	29	69	
FLUSHING	0	167	0	2	3264	454	
FOREST HILLS	0	72	0	4	797	60	
FRESH MEADOWS	0	59	0	0	604	296	
GLEN OAKS	0	5	0	0	42	49	
HOLLIS	0	37	0	0	385	146	
HOWARD BEACH	0	46	0	1	189	148	
Howard Beach	0	0	0	0	1	0	

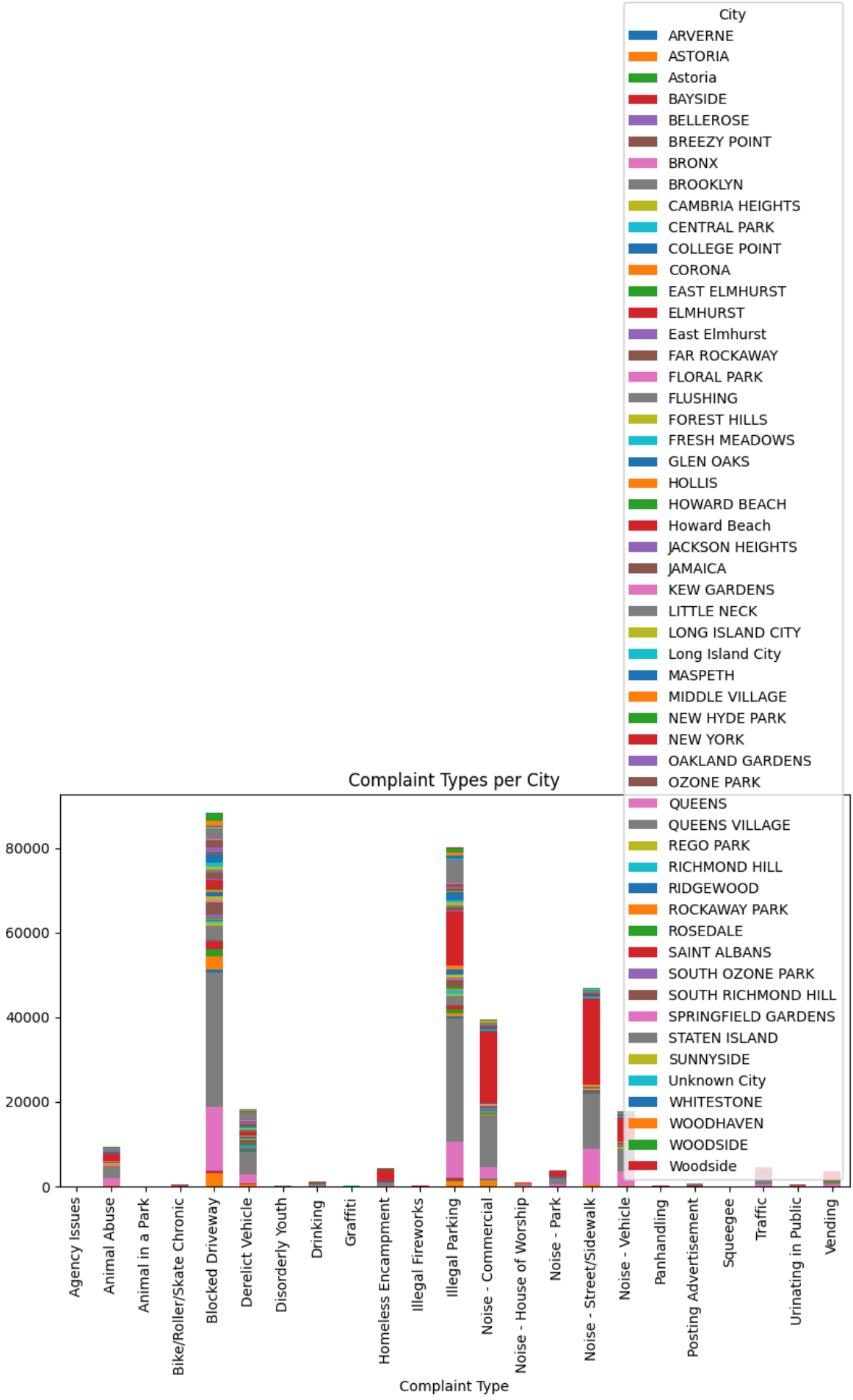
Complaint Type City	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorde You
JACKSON HEIGHTS	0	49	0	2	618	29	
JAMAICA	0	268	0	3	3057	941	
KEW GARDENS	0	25	0	0	373	15	
LITTLE NECK	0	18	0	0	155	65	
LONG ISLAND CITY	0	38	0	3	829	194	
Long Island City	0	0	0	0	45	4	
MASPETH	0	47	0	0	891	439	
MIDDLE VILLAGE	0	32	0	1	607	302	
NEW HYDE PARK	0	1	0	0	67	9	
NEW YORK	0	1723	0	228	2374	577	
OAKLAND GARDENS	0	24	0	2	157	95	
OZONE PARK	0	63	0	1	1462	430	
QUEENS	0	1	1	0	3	2	
QUEENS VILLAGE	0	78	0	0	664	419	
REGO PARK	0	29	0	0	703	79	
RICHMOND HILL	0	50	0	0	973	176	
RIDGEWOOD	0	139	0	3	1933	426	
ROCKAWAY PARK	0	33	0	0	72	15	
ROSEDALE	0	36	0	2	237	215	
SAINT ALBANS	0	38	0	0	264	205	
SOUTH OZONE PARK	0	64	0	1	1070	368	
SOUTH RICHMOND HILL	0	35	0	1	1732	314	

Complaint Type City	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorde You
SPRINGFIELD GARDENS	0	37	0	0	294	221	
STATEN ISLAND	0	691	0	10	2518	1914	
SUNNYSIDE	0	37	0	2	251	11	
Unknown City	7	1	0	2	70	52	
WHITESTONE	0	39	0	4	262	233	
WOODHAVEN	0	52	0	2	1214	322	
WOODSIDE	0	97	0	5	1796	254	
Woodside	0	0	0	0	27	8	

54 rows × 23 columns

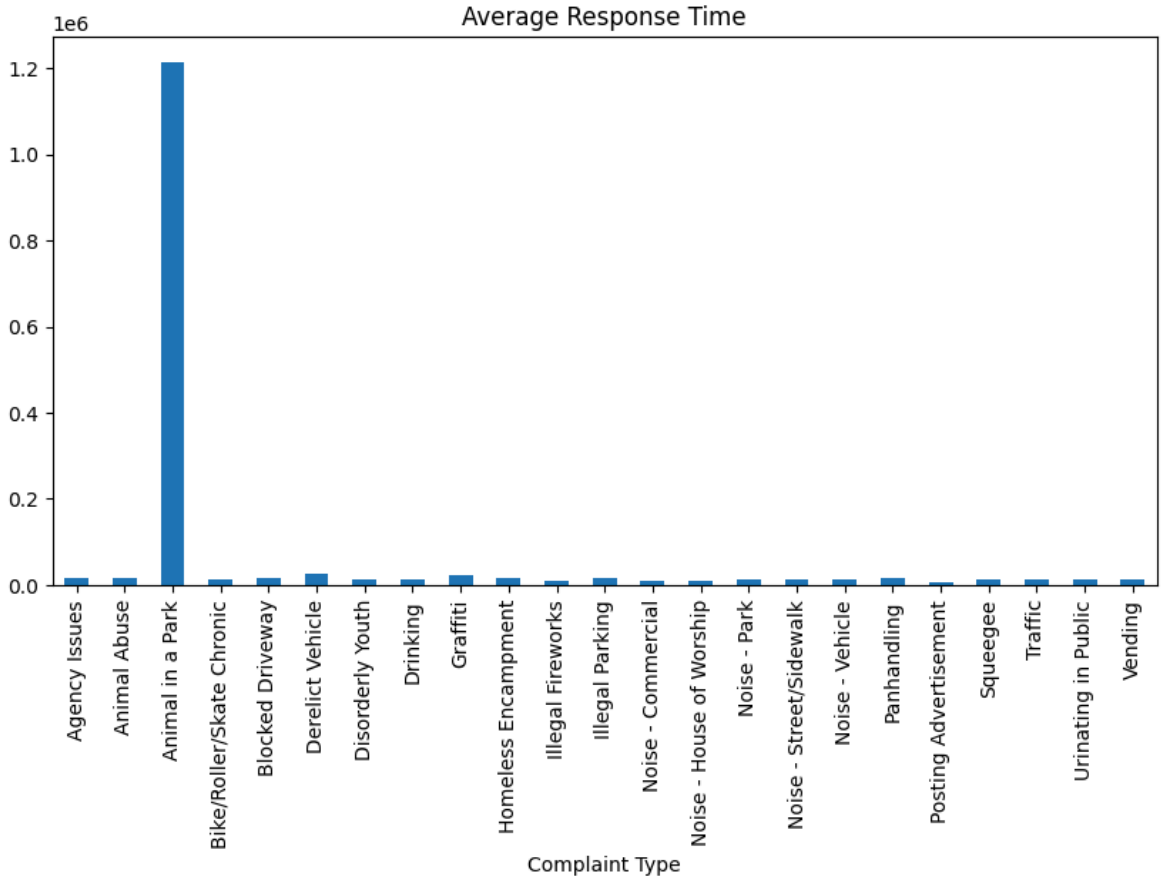
visualize the major types of complaints in each city

In [28]:
```
# Draw another chart that shows the types of complaints in each city in a single
df_new = pd.crosstab(df['Complaint Type'], df['City'])
df_new.plot(kind='bar', stacked=True, figsize=(10,5))
plt.title("Complaint Types per City")
plt.show()
```

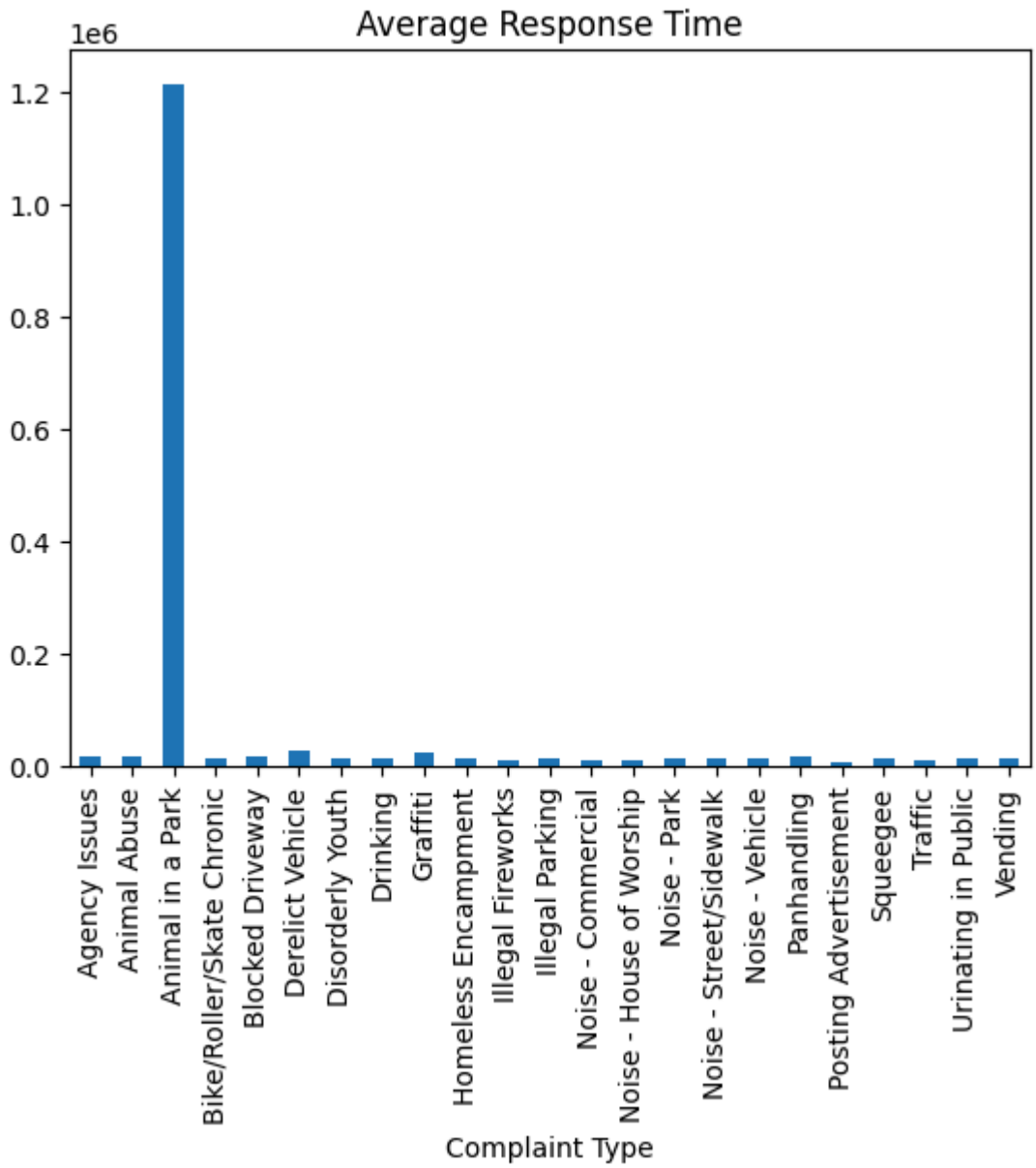


In [29]:
```
# Sort the complaint types based on the average Request_Closing_Timegrouping the
avg_time = df.groupby(['Complaint Type'])['Request_Closing_Time'].mean()
```

```
avg_time.plot(kind='bar', figsize=(10,5))
plt.title("Average Response Time")
plt.show()
```



In [30]:
```
# Visualize the average of Request_Closing_Time
df.groupby('Complaint Type')['Request_Closing_Time'].mean().plot(kind='bar')
plt.title("Average Response Time")
plt.show()
```



In [31]:
```
from scipy import stats
```

In [38]:
```
### Identify the significant variables by performing statistical analysis using
top2 = df['Complaint Type'].value_counts().index[:2]

type1 = df[df['Complaint Type'] == top2[0]]['Request_Closing_Time']
type2 = df[df['Complaint Type'] == top2[1]]['Request_Closing_Time']
t_stat, p_value = stats.ttest_ind(type1, type2, nan_policy='omit')

print("P-value:", p_value)

if p_value < 0.05:
    print("Significant difference")
else:
    print("No significant difference")
```

```
P-value: 1.5316086252342656e-08
Significant difference
```

Perform a Kruskal-WallisH test

In [39]:
```
from scipy.stats import kruskal

top_types = df['Complaint Type'].value_counts().head(5).index

groups = [df[df['Complaint Type']==t]['Request_Closing_Time'] for t in top_types

stat, p = kruskal(*groups)

print("p-value:", p)

if p > 0.05:
    print("Fail to reject H0 (Same distribution)")
else:
    print("Reject H0 (Different distribution)")
```

```
p-value: 0.0
Reject H0 (Different distribution)
```

Observation

In [35]:
```
# Most complaints are reported from New York City.
```

In [36]:
```
# Missing values were found in some columns like City and Complaint Type.
#These were handled properly (e.g., replacing with "Unknown City").
```

In [37]:
```
#Bar graphs and plots clearly show that certain complaint types dominate.
#Scatter and hexbin plots show high complaint concentration in Brooklyn
```

In [ ]:
```
#Statistical testing (Kruskal-Wallis test) was performed.
#The p-value indicates whether the response time differs across complaint types.
#Based on p-value:If p < 0.05 → There is a significant difference  If p > 0.05 →
```